

Part I

SUMMARY of Derivatives of Inverse Functions

Summary

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$$\begin{aligned}
 & \bullet \frac{d}{dx} [\sin^{-1}(x)] = \frac{1}{\sqrt{1-x^2}} & \bullet \frac{d}{dx} [\cos^{-1}(x)] = \frac{-1}{\sqrt{1-x^2}} \\
 & \bullet \frac{d}{dx} [\tan^{-1}(x)] = \frac{1}{1+x^2} & \bullet \frac{d}{dx} [\cot^{-1}(x)] = \frac{-1}{1+x^2} \\
 & \bullet \frac{d}{dx} [\sec^{-1}(x)] = \frac{1}{|x|\sqrt{x^2-1}} \text{ for } |x| \geq 1 & \bullet \frac{d}{dx} [\csc^{-1}(x)] = \frac{-1}{|x|\sqrt{x^2-1}} \text{ for } |x| \geq 1
 \end{aligned}$$

Theorem 1 (The Inverse Function Theorem). If f is a differentiable function that is one-to-one near a and $f'(a) \neq 0$, then:

- (a) $f^{-1}(x)$ is **defined** for x near $b = f(a)$,
- (b) $f^{-1}(x)$ is **differentiable** for x near $b = f(a)$,
- (c) last, but not least: $\left[\frac{d}{dx} f^{-1}(x) \right]_{x=b} = \frac{1}{f'(a)}$ where $b = f(a)$.

Part II

Recitation Questions

Problem 1

February 21, 2026

Problem 1

Problem 1 *Explain what each of the following means:*

(a) $\sin^{-1}(x)$

(b) $(\sin(x))^{-1}$

(c) $\sin(x^{-1})$

Problem 2

February 21, 2026

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Problem 2 Without using a calculator, determine if the statement below is true or false.

$$\cos^{-1}(\cos(7\pi/6)) = 7\pi/6$$

Problem 3

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Problem 3

Problem 3 *True or False:* $\sin^{-1}(0) = \pi$.

Problem 4

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Problem 4

Problem 4 *Simplify each of the following expressions.*

(a) $\cos^{-1}(\sin(\pi/2))$

(b) $\tan(\sin^{-1}(x/4))$

Problem 5

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Problem 5

Problem 5 A table of values for f and f' is shown below. Suppose that f is a one-to-one function and f^{-1} is its inverse.

x	$f(x)$	$f'(x)$
1	3	4
3	4	5
4	6	3

- (a) Evaluate $f^{-1}(f(x))$ at $x = 3$.
- (b) Evaluate $\frac{d}{dx}f(f(x))$ at $x = 3$.
- (c) Evaluate $\frac{d}{dx}\ln(f(x))$ at $x = 3$. [2]
- (d) Evaluate $f^{-1}(x)$ at $x = 3$. [2]
- (e) Evaluate $\frac{d}{dx}f^{-1}(x)$ at $x = 3$. [2]
- (f) Evaluate $\lim_{x \rightarrow 4} \frac{f(x) - f(4)}{x - 4}$ [2]

Problem 6

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Problem 6

Problem 6 *An object is moving along a horizontal line. Its position (in meters) at the time t (in seconds) is given by $s(t)$. What does the value $s^{-1}(5)$ represent?*

Problem 7

February 21, 2026

Problem 7

Problem 7 Given the expression for $f(x)$, find the derivative of f^{-1} at the given point on the graph of f^{-1} , without solving for f^{-1} .

- (a) $f(x) = x^2 + 1$ (for $x \geq 0$); the point on the graph of f^{-1} : $(5, 2)$.

Verify your answer by evaluating the derivative of f^{-1} at the given point.

- (b) $f(x) = x^2 - 2x - 3$ (for $x \leq 1$); the point on the graph of f^{-1} : $(12, -3)$.

Problem 8

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Problem 8

Problem 8 Find the slope of the tangent line to the curve $y = f^{-1}(x)$ at $(4, 7)$ if the slope of the tangent line to the curve $y = f(x)$ at $(7, 4)$ is $\frac{2}{3}$.

Problem 9

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Problem 9 Find the derivatives of the following functions:

(a) $f(x) = \sec^{-1}(\sqrt{x})$.

(b) $g(x) = \ln(\sin^{-1}(x))$.

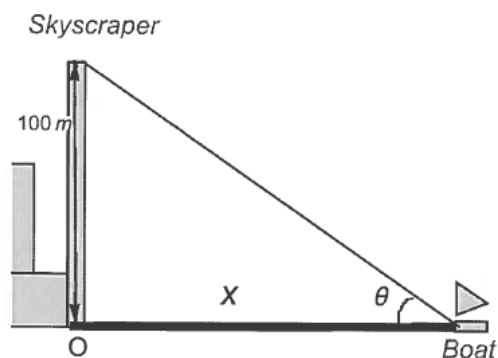
(c) $h(x) = \frac{1}{\tan^{-1}(x^2 + 4)}$.

Problem 10

February 21, 2026

Problem 10

Problem 10 A boat sails directly toward a 100-meter skyscraper that stands on the edge of a harbor. The angular size θ of the building is the angle formed by lines from the top and bottom of the building to the observer on the boat (see figure below).



- Express the angle θ as the function of x , the distance of the boat from the building.
- The boat is sailing directly toward the skyscraper at 3 m/s. Find $\frac{d\theta}{dt}$ when the boat is $x = 300\text{m}$ from the building. [3]